

Student Takeaway for Sync Session

<<Week 12: Paradigms of Machine Learning – Part II >>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

Session title	Paradigms of Machine Learning – Part II	
Session duration	1 hour 45 minutes	
Session type	Lectures: Conceptual understanding of learning paradigms (supervised, self-supervised, few-shot, federated, and transfer learning), strategic choices and trade-offs, and paradigm applications. Case Studies: Applications in healthcare, multilingual systems, and smart IoT devices with demo-based learning.	
Scope	This session explores different learning paradigms and how they influence strategy and efficiency in real-world AI/ML applications. It covers: - Supervised, self-supervised, few-shot, zero-shot, transfer, and federated learning approaches. - Factors affecting paradigm choice (data availability, privacy, compute). - Paradigm trade-offs and pitfalls. - Real-life case studies in health, language, and smart edge applications.	
Learning objectives	Objective	Core capability
	Understand visual and speech data intelligence	Real-world awareness of AI/ML application
	Apply image/video classification and captioning	Ability to use deep learning tools and models
	Explore speech recognition and synthesis	Practical understanding of ASR and TTS systems
	Connect AI/ML tools with use cases	Problem-solving in multimedia contexts

Pro tips for success:

Ask Strategic Questions: Consider data availability, compute limits, and privacy.

Relate to Cases: Match paradigm features to scenario needs.

Compare Paradigms: Think about trade-offs before choosing.

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1-2	Why Strategy Matters in ML	Importance of planning and efficient resource use	Key to project success
3	Constraints That Affect ML Decisions	Factors like data, privacy, compute	Sets the stage for paradigm choice
4-5	Learning Paradigms in AI	Overview of supervised, self-supervised, transfer, few-shot, federated learning	Foundation for rest of session
6	Important Questions to Consider	Key decision-making factors (labels, domain shift, speed)	Helps evaluate paradigm fit
7	Paradigm Trade-offs	Label need, compute cost, privacy fit	Understand what each approach demands
8	Learning Paradigms at a Glance	Quick comparison chart	Helps synthesize choices
9	Decision-Making Steps	Share needs → explain choice	Framework to apply in real cases
10-11	Case 1: Rural Health Assistant	Choose best approach under privacy and compute constraints	Transfer, federated, few-shot discussed
12-13	Case 2: Multilingual News Classifier	Handle low-resource language tasks	Explore zero-shot, translate-supervised options
14-15	Case 3: Classifier for Remote Devices	Image detection with limited resources	Compare self-supervised and transfer learning
16-20	Demo Setup: Feedback Classifier	Code walkthrough: pretrained embeddings, fine-tuning, federated training	Realistic ML pipeline
21	Output Review and Discussion	Accuracy vs. privacy outcomes	Centralised vs. federated comparison
22	AI Toolkit Cheat Sheet	Frameworks mapped to paradigms	Quick reference for your own projects
23	Paradigm-to-Constraint Matrix	Visual summary of fits and constraints	Helpful for decision logic

Slide(s)	Topic/Issue	What to expect	Notes
24–28	Common Pitfalls	Misuse of paradigms, incorrect assumptions	Helps avoid costly mistakes
29–30	General Pitfalls	Ignoring constraints, buzzword choices	Critical thinking alert
31	Impact of Paradigm Pitfalls	Project failure, resource waste	Emphasizes informed decisions
32	Best Practices	Validation, hybrid approaches	Grounded recommendations
33–36	Real-World Snapshots	Industry examples across paradigms	Google Gboard, GPT, medical AI, DINO
37	Q&A	Open floor for clarification	Wrap-up

Post session

Reflection challenge	Which paradigm would you use in your current project or context, and why?
Explore more	• Read: Case studies on AI deployment in healthcare and privacy-sensitive environments • Watch: Videos on federated learning and zero-shot NLP • Experiment: Try using a pretrained model with only 100 samples and evaluate the results
Get inspired	Strategic ML choices aren't just theoretical—they're what powers AI in remote clinics, multilingual newsrooms, smart farms, and secure mobile apps.
The journey ahead	In our next session, we'll continue with advanced AI evaluation methods and interpretability tools to help you understand and trust model outputs.